MILITARY SPECIFICATION

CIRCUIT ASSEMBLY: 11737350 (READOUT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electronic assembly known as Circuit Assembly: 11737350 (Readout) (see 6.1).

2. APPLICABLE DOCUMENTS

2. 1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification: to the extent specified herein.

SPECIFICA tions

Military

MIL-F- 13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Ins pection Equipment; Acquisition, Maintenance, and Disposition of
MIL-S-45743	Soldering, A4anual type, High llt'liability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control Systems, Procedure for

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:35Z
Check the source to verify that this is the current version before use.

MIL-S-46844	Solder Bath Solderi~g of Pr:nted Wiring Assemblies, Automatic Machine Type

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-81O	Environmental Test Methods

DRAWINGS

U.S. Army, Frankform Arsenal

F11737350	Circuit Assembly: 11737350 (Readout]

PACKAGING DATA SHEET

11737350	Packaging of Circuit Assembly; 11737350 (Readout)

(Copies of specifications, standards. drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by lke contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:35Z
Check the source to verify that this is the currentversion before use.

3.0 REQUIREMENTS

3.1 Fabrication. The circuit assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing F11737350 and drawings pertaining thereto.

3.2 General specification. The assembly shall meet the following requirements, where applicable of MIL-F-13926:

a. Order of precedence
b. Dimensions and tolerances
c. Part identification and marking
d. Electrical and electronic assemblies
e. Workmanship.

3.2.1 Soldering. Soldering shall be in accordance with MIL-S-45743 or MIL-S-46844, except that bare copper wire exposure shall be permitted at clipped ends of component leads.

3.3 Perormance. Unless otherwise specified, the assembly shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F) and 90°F.

3.3.1 Power and signals. The assembly shall perform as specified herein when the power and signals of table I are applied as specified.

TABLE I
POWER AND SIGNALS

Item	Condition	Characteristics	Connections
1.	Power sources:		Connected between the following pins
1.1	+5 Vdc	Tolerance: ±0.2 Vdc Ripple: ≤25 mV p-p	J1-1(+) and J1-3(-)
1.2	+10 Vdc	Tolerance: ±0.2 Vdc Ripple: ≤25 mV p-p	J1-2(+) and J1-3(-)
2.	Signal sources:		
2.1	Digital Type A	Logical one: +4.0 ±1.0V Source impedance: 6KΩ ±15 percent Logical zero: +0.2 ±0.2V	Applied as specified in table II herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:35Z
Check the source to verify that this is the current version before use.

3.3.1.1 Returns and range readout selection logic. with the digital Type-A signal (item 2.1) of table I applied as specified in table II, the displays shall be as specified in table II.

3.3.2 Environmental.

3.3.2.1 Storage temperature. The assembly shall meet the requirements of 3.3.1 at ambient temperature after exposure to and thermal stabilization at -80°F and +160°F.

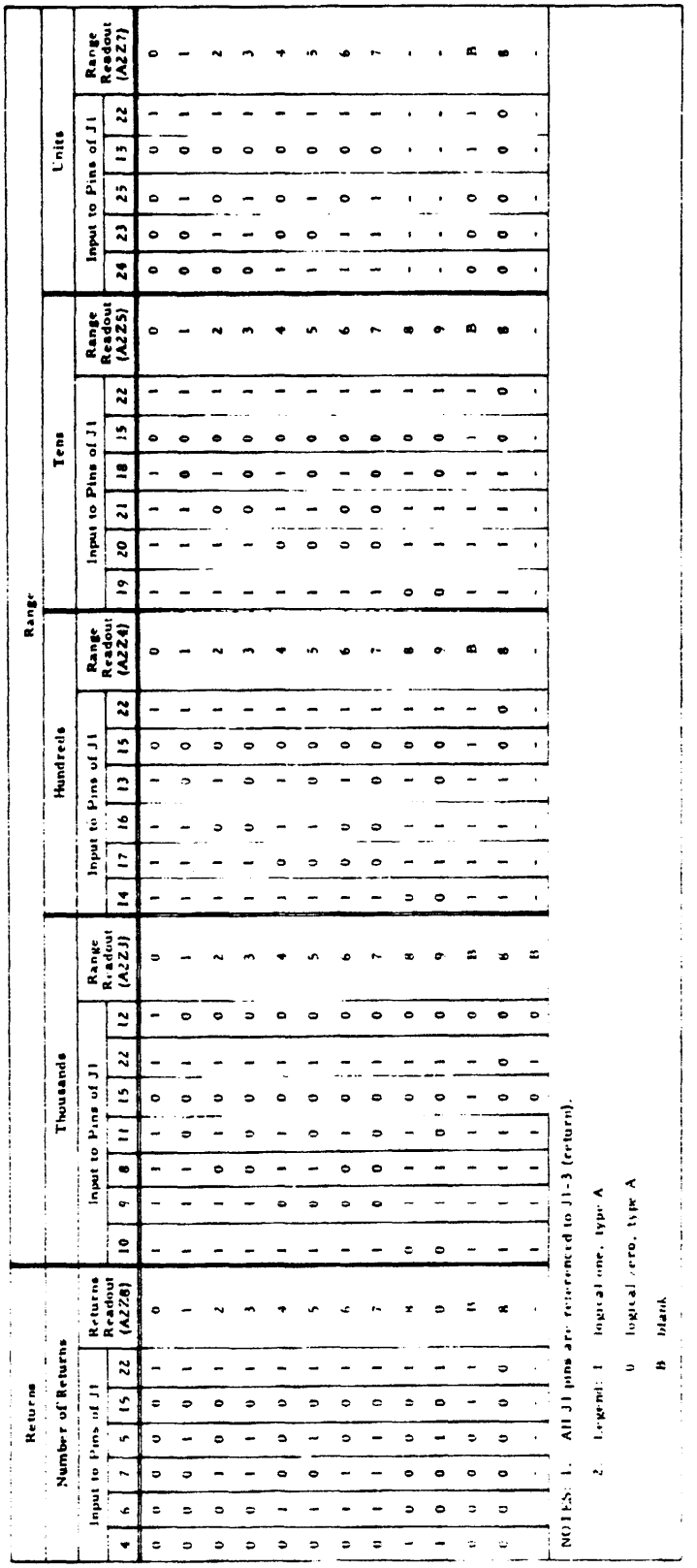
3.3.2.2 Operating temperatures. The assembly shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which the requirements of 3.3. 1 shall be met at ambient temperature.

3.3.2.3 Vibration. The assembly shall show no physical failure and shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

a. Amplitude:	1/64 inch, constant (1/32 inch double amplitude)
b. Vibratory motion:	Simple harmonic
c. Sweep time:	5 to 55 to 5 Hertz in a five-minute period
d. Vibration cycling:	Three linear or logarithmic sweeps applied to each of the three mutually perpendicular axes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:35Z
Check the source to verify that this is the current version before use.

TABLE II
RETURNS AND RANGE READOUT SELECTION LOGIC



| Returns | | |

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his won or amy other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article {initial production approval, The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three assemblies. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Governmetn testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure conformance to all the requirements of this specification. A record of this inspection, including certificates of conformance for materials, shall be submitted with the sample.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification. The sample and a record of this inspection, including certificates os conformance for materials, shall be submitted to the Government for approv'al. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified herein, or bv the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection and sampling procedures herein and submission of product provisions of MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:35Z
Check the source to verify that this is the current version before use.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspections provisions contained in the respective specifications and Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

TABLE III
FUNCTIONAL TESTS

Classification	Characteristic	Requirement	Test Paragraph
101.	Returns and range readout selection logic	3.3.1.1	4.6.2

TABLE IV
SPECIAL SAMPLING - GENERAL

Classification	Control Test	Requirement	Test Paragraph
301.	Fabrication	3.1	Applicable drawings-visual
302.	General specification	3.2	MIL-F-13926 - visual

TABLE V
SPECIAL SAMPLING - ENVIRONMENTAL

Classification	Control Test	Requirement	Test Paragraph
303.	Storage temperature	3.3.2.1	4.6.3.1
304.	Operating temperature	3.3.2.2	4.6.3.1
305.	Vibration	3.3.2.3	4.6.3.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:35Z
Check the source to verify that this is the current version before use.

4.3.2.2 Functional tests. The requirements and tests in table III shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature.

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

4.4 Special sampling. The selection of samples shall be in accordance with section 7 of MIL-STD-105.

4.4.1 General. One assembly shall be selected at random by a Government representative as a control sample from each 50 produced. The control sample shall meet the requirements and tests in table IV.

4.4.2 Environmental. Three assemblies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control sample s shall meet the requirements and tests in table V.

4.4.3 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.3.3).

4.3 Inspection equipment. Except as otherwise provided for by the contract the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. The inspection equipment listed in table VI shall be utilized in performing the tests specified herein.

TABLE VI
INSPECTION EQUIPMENT

Item	Nomenclature	Description
1.	Dc power supply	+5.0 ±0.2 V, Ripple: 25 mV p-p maximum
2.	Dc power supply	+10.0 ±0.4 V, Ripple: 25 mV p-p maximum
3.	Signal source	Capable of providing the logic levels specified in item 2.1, table I.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T20:35Z
Check the source to verify that this is the current version before use.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished inspection equipment, supply and maintenance of inspection equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment. specified by drawing number in specifications or SQAPS forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4. 5. 2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the test methods and procedures specified in 4.6 of this specification and with the component inspection procedures specified in examination and test facilities requirements of MIL-F-l3926. Since tolerance of inspection equipment is normally considered to "be within 10 percent of the pruduct tolerance for which intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept construction, materials, dimensions, and tolerances used in the design of inspection equipment shall be so selected and controlled as to insure that the inspection equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of inspection equipment.

4.6 Test methods and procedures.

4.6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MIL-F-13926.

4.6.1.1 Initial conditions. Connect the power to the assembly as specified in table I.

4. 6. 2 Returns and range readout selection logic. Apply the initial conditions of 4.6.1.1, apply the digital Type-A signal of table I of the logic level indicated, in turn, to the group of terminals associated with each readout specified in table II. Observe the readout number on the display to determine compliance with 3.3.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:35Z
Check the source to verify that this is the currentversion before use.

4.6.3 Environmental tests.

4.6.3.1 Temperature. The test apparatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926 except that the assembly may be mounted on the inside surface of the test chamber. Perform this test in accordance with Procedure I of MIL-F-13926, with conditions C (-40°F) and E (+125°F), and with the rate of temperature change during cycling not greater than 8°F per minute. At each of the specified operating temperatures, perform the test of 4.6.2 to determine compliance with 3.3.2.1 and 3.3.2.2.

4.6.3.2 Vibration. Perform this test utilizing apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Part 1 of Procedure I therin, except that the vibratory conditions of 3.3.2.3 shall apply. Following vibration exposure, examine the assembly for structural failures, loose hardware, wire breadage, and abrasion. Perform the test of 4.6.2 to determine compliance with 3.3.2.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:35Z
Check the source to verify that this is the current version before use.

5.0 PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11737350. The level of protection shall be as specified in the procurement document.

6.0 NOTES

6.1 Intended use. The circuit assembly is a functional module of the Control, Commander's, Laser Fanging C-9134()VVG-1, intended for use in the Fire Control Rangefinder AN\VVG-1(). The assembly contains readout devices which display the number of range returns. target range, and malfunction numbers.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification

b. Applicable packaging data sheet number (see 5. 1)

c. Selection of applicable levels of preservation, packaging and packing

d. Applicable stock number

e. Requirement for First Article Submission.

Custodian:

Army - MU

Preparing Activity:

Army - MU

PROJECT NO. 1240-A304

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:35Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-50736(MU) Circuit Assembly: 11737350 (Readout)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:35Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Philadelphia, Penna. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:35Z
Check the source to verify that this is the current version before use.